

Homology modeling of membrane proteins: A critical assessment

Ch. Surendhar Reddy, K. Vijayasathya, E. Srinivas, G. Madhavi Sastry*, G. Narahari Sastry*

Molecular Modelling Group, Organic Chemical Sciences, Indian Institute of Chemical Technology, Tarnaka, Hyderabad 500007, India

Received 27 August 2005; received in revised form 10 November 2005; accepted 14 December 2005

Abstract

Evaluation and validation of homology modeling protocols are indispensable for membrane proteins as experimental determination of their three-dimensional structure is an arduous task. The prediction ability of Modeller, MOE, InsightII-Homology and Swiss-PdbViewer (SPV) with different sequence alignments CLUSTALW, BLAST and 3D-JIGSAW have been assessed. The sequence identity of the target and template was chosen to be in the range of 25–35%. Validation protocols to assess the structure, fold and stereochemical quality, are employed by comparing with experimental structures. Two different ranking schemes are suggested to evaluate the performance of each methodology based on the validation scores. While unambiguous preference for any given procedure did not surface, statistically Modeller and the sequence alignment technique, 3D-JIGSAW, gave best results amongst the chosen protocols. The present study helps in selecting the right protocols when modeling membrane proteins, which form a major class of drug targets.

© 2005 Elsevier Ltd. All rights reserved.

Keywords: Membrane proteins; 3-Dimensional structure; Homology modeling; Low sequence identity; Stereochemical quality; Structural similarity; Fold

1. Introduction

Homology modeling procedures are indispensable tools for conducting research involving structure based drug design when the experimental 3D-structure of the receptor is not available (Sanchez and Sali, 1997). This has resulted in large-scale critical assessments of the performance of various automated structure prediction servers (Bujnicki et al., 2001). Although several procedures exist in the public domain, the choice of the right method is not unambiguous (Wallner and Elofsson, 2005). Therefore, it is extremely important to persistently evaluate and validate these methods especially when applied to membrane proteins due to the difficulty in obtaining their experimental 3D-structures. This is amply evident from a recent analysis of the protein databank (PDB) (Berman et al., 2000), obtained from the PDBbeta site, which contain only 1526 membrane protein structures (about 2% of total) are available as of now (July 2005). However, many of these structures belong to the water soluble domains (Tusnády et al., 2005). The recent increase in the number of crystal structures of membrane proteins has enabled an analysis of the hydrophilic exposed surfaces and the interiors of such molecules on the basis

of structure, rather than sequence alone. In spite of the increase in the number of crystal structures many structures having low resolution and high *R* factors exist in the PDB. More than 46% and 10% of the membrane protein structures have resolution ranging between 2–3 and 3–4 Å, respectively, which collectively constitute more than 56%. Therefore, homology modeling techniques are extremely important in obtaining the 3D-structure as well as in refining the existing low accuracy experimental structures (Sanchez et al., 2000). However, various protocols including sequence alignment algorithms, structure building protocols, loop building algorithms, etc., determine the accuracy of the models and choosing the right protocol is imminent to obtain good quality models (Mosimann et al., 1995).

In this work, we would like to assess various procedures when applied to model membrane proteins. We have attempted to assess the methods and servers, which are not restricted to modeling any specific class of proteins. The chosen methods and sequence alignment techniques have varied *modus operandi* of 3D model building adopted by them. The methods that do not produce any breaks in the 3D models built by them and modeled most proteins are chosen for comparison. Thus, we have considered the following methods implemented by: Modeller (Sali and Blundell, 1993), MOE (Kelly, 1999), homology module of InsightII (IH) (Dayringer et al., 1986), InsightII-Homology + Modeller (IHM) and Swiss-PdbViewer (Schwede et al., 2003; Guex and Peitsch, 1997). The central dogma in

* Corresponding authors. Tel.: +91 40 2716 0123x2619; fax: +91 40 2716 0512.

E-mail address: gnsastry@iict.res.in (G.N. Sastry).

homology modeling techniques is the right choice of alignment between sequences. The sequence alignment techniques that differ substantially from each other have been chosen for implementation, which include CLUSTALW (Thompson et al., 1994), 3D-JIGSAW (Bates et al., 2001; Contreras-Moreira and Bates, 2002) and BLAST (Altschul et al., 1990, 1997).

Different classes of membrane proteins are chosen from the PDB for modeling, to ensure diversity. The real challenge is to model structures having low sequence identities with the templates available in the PDB, but which are well within the limits of obtaining reasonably good models. For modeling membrane proteins this is a challenge, since most of these proteins have low sequence identity with the structures available in PDB (Berman et al., 2000). Therefore, template structures whose percentage identity ranges between 25% and 32% with the target sequence whose 3D-structure is known have been chosen to build the models. The 3D models obtained for the target sequence, are compared with the original PDB structure and validated by implementing varied protocols which include root mean square deviation (RMSD) (Carugo and Pongor, 2001), self-compatibility scores (SCS) (Luthy et al., 1992; Eisenberg et al., 1997), template modeling (TM) score (Zhang and Skolnick, 2004) and Ramachandran validation (RC) (<http://raven.bioc.cam.ac.uk/rampage.php>). Different combinations of the abovementioned methods and sequence alignments lead to 17 methodologies and the built models were subsequently subjected to structural assessment. The whole process was repeated for 20 proteins and the results are generalized to understand the performance of the methodologies. The PDB codes of the considered test and template proteins, together with the structural alignment, number of residues present in loops, self-compatibility scores of the individual proteins are provided in the supporting information. It should be noted that very few structures of membrane proteins are present in the PDB which have a homologous counterpart in the database having high crystallographic resolution and low *R*-factor. The term methodology will be used for the combination of a method and sequence alignment technique throughout the manuscript. The methodologies are finally ranked according to each validation protocol wherein two different ranking schemes are proposed and implemented.

2. Materials and methods

At the outset, crystal structures of membrane proteins belonging to different classes are selected from the PDB so as to ensure diversity. A homologous protein in the PDB having sequence identity between 25% and 32% with the target sequence is identified first. The resolution and *R*-value of the templates selected from PDB was less than 3.0 Å and 0.3, respectively. Care was taken to choose target and template structures which have good structural alignments with low RMSD values. Structural comparison between the target and template protein structures is made to gauge the intricacy of the task (see [supporting information](#)). The RMSD values between the target and the template are by and large below 3 Å. Pairs of proteins with variable total loop lengths have been chosen for modeling. Since, the difference in loop lengths plays a major role in protein modeling,

we have evaluated the total number of residues present in the loops of the target and the template. Although, the maximum difference in the number of loop residues is 38, the RMSD is quite low (1.2 Å). In contrast, the target–template pair with the highest RMSD has a difference of only 10 loop residues. Thus, the difference in loop lengths do not seem to have any direct correlation with the structural alignment, at least in the chosen data set. In all cases, the SCS values of the target as well as the templates are well within the accuracy limits indicating the reliability of the fold description of the models obtained based on these PDB structures.

The chosen methods adopt varied protocols to fabricate the 3D models of the proteins. Modeller implements comparative protein structure modeling by satisfaction of spatial restraints and performs optimization of various models of protein structure with respect to a flexibly defined objective function (Sali and Blundell, 1993). The method IH, which mostly depends on human interface, builds the model depending on target and template alignment wherein the loops are built using de novo prediction using random conformational searches based on energy criteria. The side chain conformations are optimized using a rotamer library (Dunbrack, 2002). The model created by IH (<http://www.accelrys.com/insight/>) is further refined by Modeller (IHM) and compared with the initial model. The method implemented by SPV uses constraint space programming (CSP) to model insertions or deletions in the target–template alignment after modeling the core, based on averaging the backbone atom position of the template (Schwede et al., 2003; Guex and Peitsch, 1997). The best loop is built using a force field and interaction energy based scoring scheme and the side chain conformations are selected based on the rotamer library and a scoring function (Dunbrack, 2002). MOE builds the initial partial geometry based on conserved regions and treats insertions and deletions by using a specialized logic (Needleman and Wunsch, 1970). The loops are initially modeled randomly and subsequently, a contact energy function is implemented to identify the possible candidates which are chosen based on Boltzmann weighted averaging (Sippl, 1993a,b).

The chosen sequence alignment techniques differ substantially from each other. BLAST optimizes a measure of local similarity termed the maximal segment pair (MSP) score (Altschul et al., 1990) whereas CLUSTALW implements a progressive multiple sequence alignment method that is improved to align divergent protein sequences (Thompson et al., 1994). On the other hand, 3D-JIGSAW implements PSI-BLAST (Altschul et al., 1997) and uses position specific scoring matrix (PSSM) of the target sequence to predict secondary structure using PSI-Pred (Jones, 1999). The target PSSM files and the template are fed into a dynamic programming algorithm to obtain the best alignment based on a secondary structure based scoring function (Bates et al., 2001).

The models were validated using different protocols. The stereochemical quality of the modeled proteins is assessed from Ramachandran validation score for favored regions and allowed regions (Lovell et al., 2003). In general, a score close to 100% implies good stereochemical quality of the models. The structural similarity between the model and the real structure from

the PDB is assessed from the RMSD values and TM score (Carugo and Pongor, 2001; Zhang and Skolnick, 2004). The RMSD approach has its attraction in the physical significance that it provides. After the optimal alignment of the structures, the RMSD values are given by Eq. (1).

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^N d_i^2}{N}} \quad (1)$$

where d_i is the distance between the i th atom of the two conformations and N is the number of such distances (Bindu et al., 2004). In general, an RMSD value, which is less than 3 Å, implies a fairly good similarity between the structures. Evaluation of the accuracy of the 3D models is a very difficult task. Although the RMSD protocol is most popularly implemented, it suffers from a few drawbacks. For example, a badly modeled part of the protein can significantly affect its values with no information on the well modeled part. Moreover, large size of the structures can result in high RMSD values. However, the TM score has no bias towards the target protein's length due to normalization that is dependent on protein size. The score is a variation of Levitt–Gerstein score (Levitt and Gerstein, 1998) used for sequence independent structure alignments. It is given by

$$\text{TM score} = \text{Max} \left[\frac{1}{L_N} \sum_1^{L_T} \frac{1}{1 + \left(\frac{d_i}{d_0}\right)^2} \right] \quad (2)$$

L_N is the native structure length, L_T the length of aligned residues to the template structure, d_0 the normalizing constant and d_i is the distance between the i th pair of aligned residues (Zhang and Skolnick, 2004). Generally, TM score discriminates a good model from an inaccurate model. A TM score close to 1 and RMSD close to 0, evaluated between the modeled structure and the real structure (from the PDB) indicate structurally good models. The folds of the models are assessed from the procedure of Luthy et al. (1992), which measures the compatibility of a protein model with its sequence. They evaluated the 3D–1D self-compatibility scores for structures in the PDB and made a log–log scatter plot of these scores against sequence length L . It was found that they fell approximately on a straight line. A linear least-squares fit to these data, yielded the following relation:

$$S_{\text{calc}} = \exp(-0.83 + 1.008 \ln(L)) \quad (3)$$

S_{calc} is the score expected for a correct structure. Severely misfolded structures typically exhibit scores less than $0.45S_{\text{calc}}$. A high scoring model indicates that the folds are well reproduced. Computing averages over the dataset, the ranking of each method or sequence alignment is done for every validation protocol. Two different ranking schemes are employed which will be detailed in the ensuing discussion. Whenever, equal values were encountered the next corresponding rank is assigned for the models by skipping the current rank. Whenever a model was not built, the corresponding procedure is assigned the last rank. It may be noted that at first glance averaging the SCS score may not seem meaningful since every protein has a different score.

Nonetheless, a large SCS indicates good folding properties for any protein and therefore this information is not lost during the averaging process.

The reliability of the methodologies are assessed by ranking the scores obtained from the validation protocols. While ranking, the numerical values of the scores have to be taken into consideration. For some proteins, the methodologies may not give very good models quantitatively according to a certain validation protocol, rendering the ranking methodologies grossly qualitative in such cases. Further, the extent of numerical differences have to be taken into account while ranking. For example, consider only two methodologies A and B. The difference in the validation scores obtained by modeling with A could be significantly higher than B in one case and marginally higher in another case. However, in both the situations A and B will be ranked as 1 and 2, respectively, which is independent of the numerical differences in scores. To develop a more dependable approach, two different ranking schemes are proposed. In the first case, each methodology is ranked according to a validation protocol for a modeled protein, and subsequently the ranks are averaged over the dataset and are finally ranked again. Mathematically, let m_i , P_k and VP_l represent the i th methodology, k th protein and l th validation protocol, respectively. Then the set $\{S(m_i), \forall i\}$ contains the scores for each methodology for a given protein P_k and VP_l . These scores are ranked given by $r(m_i, P_k)$ for a given methodology and validation protocol. Subsequently, averaging over the dataset for a methodology is performed, i.e.,

$$\left[\frac{1}{N} \sum_{k=1}^N r(m_i, P_k) \right]_{VP_l} = v_i \quad (4)$$

N is the number of proteins in the dataset and is equal to 20 in this study. The set of values $\{v_i, \forall i\}$ is finally ranked again for a given protocol VP_l . In the second ranking scheme, the validation scores are first averaged over the dataset for a particular methodology. Subsequently, the averages obtained for each methodology are finally ranked. Let $S(P_k)$ represent the score of a protein for a given m_i and VP_l . Then $\bar{S}_i$ represents the average score over the dataset for a given protocol and methodology, which is given by

$$\bar{S}_i = \left[\frac{1}{N} \sum_{k=1}^N S(P_k) \right]_{m_i, VP_l} \quad (5)$$

The set $\{\bar{S}_i, \forall i\}$ is finally ranked for a given VP_l .

The former scheme which we denote as RA, gives information on which method performs better statistically by considering its performance in each individual case. The difference in the numerical values of the validation scores obtained from the different methodologies does not affect the RA rank. In contrast, differences in numerical values of the different scoring protocols influence the latter ranking scheme which we denote as AR. In general, a method having high RA (*low numerical value*) indicates that its performance is best in most cases. A method having high AR but low RA indicates its extremely good performance relative to the others in a few cases and vice versa. However, a very bad model can influence the final AR scoring by low-

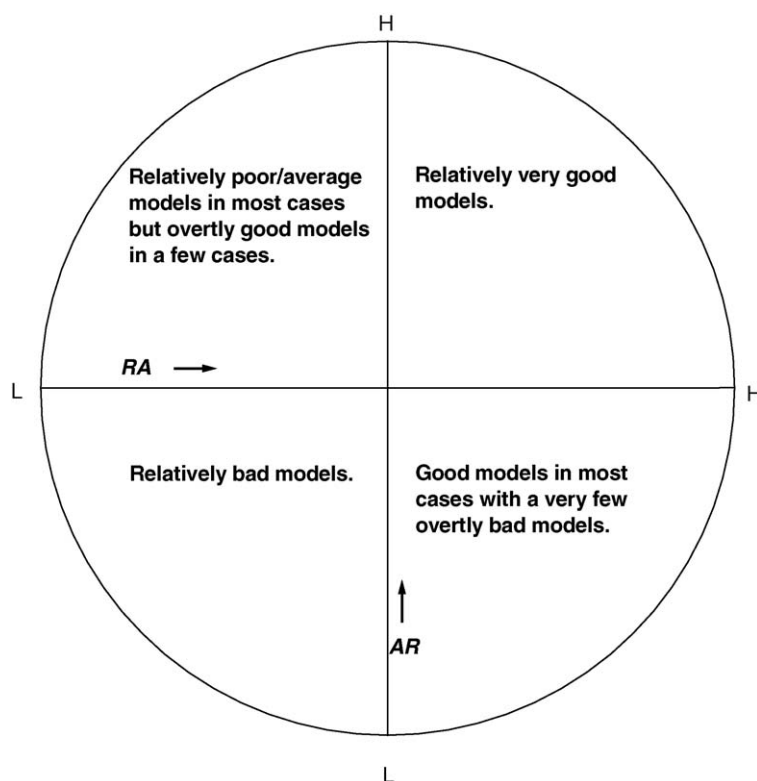


Chart 1. Statistical interpretation of the combination of the two ranking protocols. H, high and L, low.

ering the score unless the other models are well reproduced. This problem does not arise in the case of RA scoring which, however, does not take the numerical differences in the validation scores into account. Interpretation of the performance of a methodology according to both RA and AR scores is given in Chart 1.

3. Results and discussion

The total number of best final models that can in principle be generated from 17 methodologies for the 20 target proteins amounts to 340. The methodologies that do not produce any breaks in the 3D models built by them are chosen for comparison. However, the subsequences at the terminals of the proteins to be modeled in most cases did not have any counterpart in the template sequence from the different sequence alignments. It should be noted that this amounted to missing only a few residues at the terminals, which were not modeled. This is also due to the different sequence lengths of the proteins. Therefore, the maximum part of the proteins that were modeled by all the procedures was considered for final comparison. The TM scores were obtained by considering the length of the protein present in the PDB.

The 17 different methodologies were systematically applied on all the model systems considered in the study. In the ensuing discussion, we will assess the prediction ability of the different methodologies with respect each validation protocol according to the two ranking schemes AR and RA.

RMSD: About 42% of the final models gave RMSD values with the experimental structures below 3.5 Å. Modeller with

3D-JIGSAW alignment statistically gave models with relatively lowest RMSD values with the experimental structure (Table 1). This methodology ranked first according to AR as well as RA. This implies that the combined methodology performs better in accordance with the RMSD protocol, numerically as well as qualitatively. Strikingly, all methods gave lowest ranks with 3D-JIGSAW sequence alignment according to both ranking schemes (Table 1) followed by CLUSTALW with noteworthy difference in the values obtained. Since both RA and AR ranks signify the best performance of 3D-JIGSAW amongst sequence alignments (Table 2) and Modeller among methods (Table 3), spurious ranks due to a relatively very good or overtly bad scores do not prevail. The Align2D alignment implemented by Modeller, which aligns block of sequences with a block of structures using gap penalty functions, gave models with relatively large RMSD values (Sali and Blundell, 1993). It should be noted that all the models obtained from different methodologies gave larger than zero RMSD values with the experimental structures, which clearly indicates that the homology modeling techniques still have a long way to go in modeling the 3D-structures almost precisely.

SCS: In contrast to RMSD values, 90% of the final models obtained from the 17 methodologies gave SCS scores above the threshold indicating better than average fold description. Statistically, both ranking schemes indicate that the best fold description is given by IHM closely followed by Modeller and subsequently SPV, provided 3D-JIGSAW alignment is implemented (Table 1). However, Table 3 shows slightly different AR and RA ranking for Modeller and IHM indicating that the fold quality of the models obtained from these two methods is almost similar. All the sequence alignments gave best fold

Table 1
Ranking of different methodologies

Protocol	Ranking	SPV			Modeller				IH			IHM			MOE			
		Bla	Clu	Jig	2D	Bla	Clu	Jig	Bla	Clu	Jig	Bla	Clu	Jig	NW	Bla	Clu	Jig
RMSD	RA	17	13	3	15	7	5	1	16	11	6	9	8	2	15	13	10	4
	AR	16	11	3	17	10	6	1	15	8	5	14	7	2	13	12	9	3
SCS	RA	15	4	3	13	7	6	2	17	11	8	10	5	1	15	16	12	10
	AR	17	12	3	14	6	5	2	16	9	7	8	4	1	15	13	11	10
RC	RA	11	9	7	3	1	4	2	17	15	16	14	13	12	10	8	7	5
	AR	5	7	6	2	1	4	3	17	15	16	14	13	12	11	10	9	8
TM	RA	12	11	1	6	8	5	2	17	16	7	15	10	4	15	9	15	4
	AR	4	1	2	9	15	8	11	17	14	13	16	12	10	6	7	5	3
Average	RA	13.75	9.25	3.5	9.25	5.75	5	1.75	16.75	13.25	9.25	12	9	4.5	13.75	11.5	11	5.75
	AR	10.5	7.75	3.5	10.5	8	5.75	4.25	16.25	11.5	10.25	13	9	6.25	11.25	10.5	8.5	6

Bla, BLAST; Clu, CLUSTALW; Jig, 3D-JIGSAW; 2D, Align2D (Modeller), NW, Needleman–Wunsch algorithm as implemented in MOE.

Table 2
Mean ranking of different sequence alignment techniques over the methods, according to different evaluation criteria

Criterion	Ranking	BLAST	CLUSTALW	3D-JIGSAW
RMSD	RA	3	2	1
	AR	3	2	1
SCS	RA	3	2	1
	AR	3	2	1
RC	RA	3	2	1
	AR	3	2	1
TM	RA	3	2	1
	AR	3	2	1

description with IHM according to AR (Table 1). However, in the case of RA, CLUSTALW alignment performed better with SPV. This implies that the SCS of models obtained by SPV using CLUSTALW and 3D-JIGSAW sequence alignments, statistically did not differ much from each other. All the methods performed better with 3D-JIGSAW sequence alignment according to both ranking schemes (Table 2).

RC: Percentage score of more than 90 is found for about 85% models out of which 66% models have greater than 95% residues within allowed regions of the Ramachandran plot (<http://raven.bioc.cam.ac.uk/rampage.php>). Best stereochemi-

Table 3
Ranking of the homology modeling methods averaged over the sequence alignments, according to different evaluation criteria

Criterion	Ranking	SPV	Modeller	IH	IHM	MOE
RMSD	RA	3	1	5	2	4
	AR	5	1	4	3	2
SCS	RA	3	2	4	1	5
	AR	5	1	3	2	4
RC	RA	3	1	5	4	2
	AR	2	1	5	4	3
TM	RA	2	1	5	3	4
	AR	1	3	5	4	2

cal quality of the models is seen to be obtained from the methodology Modeller+BLAST closely followed by Modeller+Align2D according to AR ranking (Table 1). However, RA ranks Modeller+3D-JIGSAW as second followed by Align2D. This slight difference in the two ranking schemes for the two alignments indicates that the difference in quality of the models obtained is negligible. All sequence alignments statistically performed best with Modeller according to both ranking schemes (Table 3). In contrast, each method differed in the sequence alignment with which it performed better according to both ranking schemes (Table 1). This is due to the fact that for a particular method, all alignments gave similar RC scores. However, statistically 3D-JIGSAW was ranked high for most methods leading to best overall performance of this alignment (Table 2). The RA and AR ranks of SPV and MOE are close indicating similar stereochemical quality of the models obtained from these methods (Table 3).

TM: A TM score above 0.4 signifies meaningful prediction of the models. A score below 0.17 indicates that the model is not well reproduced and indicates random predictions. About 65% models obtained from the 17 methodologies gave values higher than 0.4 and 20% of the proteins exhibit scores less than 0.17. SPV with CLUSTALW followed by 3D-JIGSAW alignment gave models with best TM scores statistically according to AR. In contrast, SPV+3D-JIGSAW ranked first followed by Modeller+3D-JIGSAW according to the RA ranking (Table 1). Amongst methods, SPV performs clearly better than the other methods according to the AR scheme followed by MOE (Table 3) for different alignments. However, Modeller is the performer according to the RA ranking. This indicates that the performance of SPV and MOE is significantly good relative to the other methods in some cases whereas Modeller generally gives reasonable models. The performance of each method is the best with 3D-JIGSAW alignment in general according to RA. However, the AR ranking shows better performance of CLUSTALW with SPV and Modeller. This is because the difference in validation scores obtained for each sequence alignment in the case of Modeller and SPV is not significant which explains the discrepancies observed in the two ranking schemes.

Yet again, the high ranks obtained for most methods with 3D-JIGSAW alignment renders it as the best alignment scheme statistically.

Amongst all the methods, Modeller, IH, IHM and MOE, yielded answers for all the twenty proteins whereas SPV failed in 10% cases. These failures may be attributed to long length of loops or absence of the right loop in the loop library. The calculations, based on the stereochemical quality, structural similarity and fold quality according to the two ranking measures, generally show that Modeller, with 3D-JIGSAW sequence alignment, performs better for the membrane proteins compared to the other methodologies (Table 1). The better performance of Modeller has been observed in a recent study (Wallner and Elofsson, 2005).

The 3D-JIGSAW alignment seems to virtually monopolize the sequence alignment techniques, since all the methods perform well with this alignment in general (Table 2). This alignment scheme splits query sequence into domains and aligns the template using PSI-BLAST. This domain splitting scheme may be necessary to model membrane proteins since they have different secondary structure and folding properties than soluble proteins due to the effect of the lipid matrix (Bates et al., 2001; Contreras-Moreira and Bates, 2002). Moreover, secondary structural features such as α -helices and β -sheets are formed with different probabilities in membrane proteins compared to soluble proteins (Wallace et al., 1986). The better efficacy of 3D-JIGSAW alignment that uses PSSM matrix for secondary structure prediction corroborates with the observations of Bujnicki et al. (2001) wherein they found that the 3D-PSSM server was amongst the top in the case of hard targets (Bujnicki et al., 2001). Interestingly, Align2D of Modeller, which aligns block of sequences with a block of structures using gap penalty functions, does not perform well amongst the sequence alignments (Sali and Blundell, 1993). The major drawback is seen in the SCS scores and RMSD values indicating that the fold description is not well reproduced for membrane proteins using this alignment. The global alignment scheme of CLUSTALW, which uses dynamic programming algorithm ranks second followed by local alignment scheme of BLAST (Table 2). This is expected since BLAST uses a local alignment scheme in contrast to multiple alignment by CLUSTALW. However, the multiple sequence alignment of CLUSTALW is built from pairwise alignments which are global and consequently the global alignment scheme is expected to cause problems when the target and template sequences have different lengths. This is reflected in the TM scores wherein the results obtained from the two methods are almost comparable in spite of a better alignment scheme of CLUSTALW. This indicates that the two methods are of comparable ability to model full length protein as in the PDB. However, CLUSTALW gave better SCS values and consequently fold description relative to BLAST irrespective of the difference in the target–template sequence lengths. The models obtained by the CLUSTALW alignment were in general better than that of BLAST according to the RMSD criterion. Earlier studies on protein modeling indicated that PSI-BLAST aligns residues much better than CLUSTALW whereas BLAST was less efficient (Elofsson, 2002). Similar results are found in the

present modeling study of membrane proteins wherein it is found that 3D-JIGSAW which implements PSI-BLAST gives statistically better results compared to CLUSTALW and BLAST. However, to know whether the effectiveness of 3D-JIGSAW is only due to the use of PSI-BLAST algorithm or whether the use of secondary structure information is important, we have analyzed the models obtained from Geno3D online server which implements PSI-BLAST for sequence alignment. It is found that 15 out of the 20 proteins could be modeled using this server. The models gave low TM scores indicating the inability to model the full length protein as in the PDB. Although the models give good stereochemical quality, fold and RMSD values, the models obtained by 3D-JIGSAW alignment were relatively better in most cases, implicating the importance of secondary structure alignment. Understandably so, since the secondary structure and fold description of membrane proteins is different from soluble proteins due to the effect of the lipid matrix (Bates et al., 2001; Contreras-Moreira and Bates, 2002).

Earlier benchmarking studies on the homology modeling methodologies and alignments is case dependent emphasizing the importance of considering more than one protocol in different prediction tasks (Briffeuil et al., 1998; Fischer et al., 1999; Moulton et al., 1999). The present study indicates that Modeller amongst methods and 3D-JIGSAW amongst sequence alignment techniques gives statistically better results in comparison to the other procedures implemented in this work. However, it should be noted that the results are statistical and no single homology modeling program amongst the chosen methodologies is best suitable to model membrane proteins. The present study helps in choosing the right protocols when modeling membrane proteins. However, taking existing experimental data and information into consideration is an indispensable prerequisite for modeling the 3D-structure of a protein.

Acknowledgements

ES acknowledges financial support from AICTE. GMS thanks DST for financial support.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.compbiolchem.2005.12.002.

References

- Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215, 403–410.
- Altschul, S.F., Madden, T.L., Schaffer, A.A., Zhang, J., Zhang, Z., Miller, W., Lipman, D.J., 1997. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res.* 25, 3389–3402.
- Bates, P.A., Kelley, L.A., MacCallum, R.M., Sternberg, M.J., 2001. Enhancement of protein modelling by human intervention in applying the automatic programs 3D-JIGSAW and 3D-PSSM. *Proteins* 5 (Suppl.), 39–46.
- Berman, H.M., Westbrook, J., Feng, Z., Gilliland, G., Bhat, T.N., Weissig, H., Shindyalov, I.N., Bourne, P.E., 2000. The protein data bank. *Nucleic Acids Res.* 28, 235–242.

- Bindu, P.H., Sastry, G.M., Murty, U.S., Sastry, G.N., 2004. Structural and conformational changes concomitant with the E1–E2 transition in H⁺K⁺-ATPase: a comparative protein modeling study. *Biochem. Biophys. Res. Commun.* 319, 312–320.
- Briffeuil, P., Baudoux, G., Lambert, C., De Bolle, X., Vinals, C., Feytmans, E., Depiereux, E., 1998. Comparative analysis of seven multiple protein sequence alignment servers: clues to enhance reliability of predictions. *Bioinformatics* 14, 357–366.
- Bujnicki, J.M., Elofsson, A., Fischer, D., Rychlewski, L., 2001. LiveBench-1: continuous benchmarking of protein structure prediction servers. *Protein Sci.* 10, 352–361.
- Carugo, O., Pongor, S., 2001. A normalized root-mean-square distance for comparing protein three-dimensional structures. *Protein Sci.* 10, 1470–1473.
- Contreras-Moreira, B., Bates, P.A., 2002. Domain fishing: a first step in protein comparative modelling. *Bioinformatics* 18, 1141–1142.
- Dayringer, H.E., Tramontano, A., Fletterick, R.J., 1986. Interactive program for visualization and modelling of proteins, nucleic acids and small molecules. *J. Mol. Graph.* 4, 82–87.
- Dunbrack Jr., R.L., 2002. Rotamer libraries in the 21st century. *Curr. Opin. Struct. Biol.* 12, 431–440.
- Eisenberg, D., Lüthy, R., Bowie, J.U., 1997. VERIFY3D: assessment of protein models with three-dimensional profiles. *Methods Enzymol.* 277, 396–404.
- Elofsson, A., 2002. A study on protein sequence alignment quality. *Proteins: Struct. Funct. Genet.* 46, 330–339.
- Fischer, D., Barret, C., Bryson, K., Elofsson, A., Godzik, A., Jones, D., Karplus, K.J., Kelley, L.A., MacCallum, R.M., Pawlowski, K., 1999. CAFASP-1: critical assessment of fully automated structure prediction methods. *Proteins* 3 (Suppl.), 209–217.
- Guex, N., Peitsch, M.C., 1997. SWISS-MODEL and the Swiss-PdbViewer: an environment for comparative protein modelling. *Electrophoresis* 18, 2714–2723.
- Jones, D.T., 1999. Protein secondary structure prediction based on position specific scoring matrices. *J. Mol. Biol.* 292, 195–202.
- Kelly, K., 1999. 3D bioinformatics and comparative protein modeling in MOE, J. Chem. Comp. Group, autumn ed.
- Levitt, M., Gerstein, M., 1998. A unified statistical framework for sequence comparison and structure comparison. *Proc. Natl. Acad. Sci. U.S.A.* 95, 5913–5920.
- Lovell, S.C., Davis, I.W., Arendall, W.B., de Bakker, P.I., Word, J.M., Prisant, M.G., Richardson, J.S., Richardson, D.C., 2003. Structure validation by Calpha geometry: phi, psi and Cbeta deviation. *Proteins* 50, 437–450.
- Luthy, R., Bowie, J.U., Eisenberg, D., 1992. Assessment of protein models with three-dimensional profiles. *Nature* 356, 83–85.
- Mosimann, S., Meleshko, R., James, M.N.G., 1995. A critical assessment of comparative molecular modeling of tertiary structures of proteins. *Proteins: Struct. Funct. Genet.* 23, 301–317.
- Moult, J., Hubbard, T., Fidelis, K., Pedersen, J.T., 1999. Critical assessment of methods of protein structure prediction (CASP): round III. *Proteins* 3 (Suppl.), 2–6.
- Needleman, S.B., Wunsch, C.D., 1970. A general method applicable to the search for similarities in the amino acid sequence of two proteins. *J. Mol. Biol.* 48, 443–453.
- Sali, A., Blundell, T.L., 1993. Comparative protein modeling of spatial restraints. *J. Mol. Biol.* 234, 779–815.
- Sanchez, R., Sali, A., 1997. Advances in comparative protein-structure modelling. *Curr. Opin. Struct. Biol.* 7, 206–214.
- Sanchez, R., Pieper, U., Melo, F., Eswar, N., Marti-Renom, M.A., Madhusudan, M.S., Mirkovic, N., Sali, A., 2000. Protein structure modeling for structural genomics. *Nat. Struct. Biol.* 7 (Suppl.), 986–990.
- Schwede, T., Kopp, J., Guex, N., Peitsch, M.C., 2003. SWISS-MODEL: an automated protein homology-modeling server. *Nucleic Acids Res.* 31, 3381–3385.
- Sippl, M.J., 1993a. Boltzmann's principle, knowledge-based mean fields and protein folding. An approach to the computational determination of protein structures. *J. Comp. Aid. Mol. Des.* 7, 473–501.
- Sippl, M.J., 1993b. Recognition of errors in three-dimensional structures of proteins. *Proteins* 17, 355–362.
- Thompson, J.D., Higgins, D.G., Gibson, T.J., 1994. CLUSTALW: improving the sensitivity of progressive multiple sequence alignment through sequence weighing, position specific gap penalties and weight matrix choice. *Nucleic Acids Res.* 22, 4673–4680.
- Tusnády, G.E., Dosztányi, Z., Simon, I., 2005. PDB.TM: selection and membrane localization of transmembrane proteins in the protein data bank. *Nucleic Acids Res.* 33, 275–278.
- Wallace, B.A., Cascio, M., Mielke, D.L., 1986. Evaluation of methods for the prediction of membrane protein secondary structures. *Proc. Natl. Acad. Sci. U.S.A.* 83, 9423–9427.
- Wallner, B., Elofsson, A., 2005. All are not equal: a benchmark of different homology modeling programs. *Protein Sci.* 14, 1315–1327.
- Zhang, Y., Skolnick, J., 2004. Scoring function for automated assessment of protein structure template quality. *Proteins* 57, 702–710.